

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Problem Solving Assignment

**COURSE NAME: ARTIFICIAL
INTELLIGENCE AND EXPERT
SYSTEMS**

**COURSE OUTCOME COVERED
COURSE CODE: MLA01**

CO2: *Analyze and select appropriate agent strategies for different types of environments and problem settings. (BTL 3)*

Assessment Tool Weightage: 40 %

Total Marks: 100 Time: 60 Mins No. of Questions: 5

Annexure A

Problem Solving Assignment

Code of Conduct:

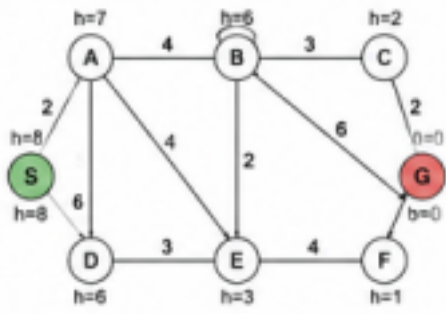
I Kamalli shree V(192525030) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Problem Analysis and Understanding	15%	
AI Algorithm Selection & Justification	15%	
Algorithm Design / Pseudocode	25%	
Application of AI Concepts (Greedy, A*, CSP, Minimax, etc.)	15%	
Diagram / State Space / Search Tree Representation	10%	
Complexity Analysis and Solution Efficiency	10%	
Originality, Critical Thinking, and Presentation Total	100%	

Signature of the Faculty Total Marks Awarded

Answer All the Questions

Q.No	Application-Based Question	Major CO2 Concepts
1	Smart Ambulance Navigation: An ambulance must travel from a hospital to an emergency location through a city road network. Some roads have higher travel costs, and an estimated distance to the destination is available for each junction. Design a search-based solution to identify the best route. Using the given road network and heuristic values, determine the route selected by your algorithm, show the sequence of junctions explored, calculate the relevant evaluation values, and justify whether the route is optimal. 	Informed Search, A*, Heuristic Function
2	Autonomous Warehouse Robot: A robot has to move from a charging station to a target shelf in a warehouse. The robot does not know all obstacles initially and discovers some obstacles only when it reaches nearby locations. Design an intelligent search strategy for the robot. Demonstrate the robot's decisions for the given grid, explain how it reacts when a new obstacle is discovered, and compare the approach with an offline search strategy. 	Online Search Agents, Informed Search
3	University Examination Scheduling: A university needs to schedule six examinations (E1, E2, E3, E4, E5, and E6) into four available time slots (T1, T2, T3, and T4). Some examinations have common students and therefore cannot be conducted in the same time slot.	CSP, MRV, Forward Checking, Constraints

	The following constraints must be satisfied: 1. E1 and E2 have common students. 2. E1 and E3 have common students. 3. E2 and E4 have common students. 4. E3 and E5 have common students. 5. E4 and E6 have common students. 6. E2 and E5 have common students. The university has three examination halls with the following capacities:	
	Examination Hall	Capacity
	Hall A	120 students
	Hall B	90 students
	Hall C	60 students
	Examination E1 E2 E3 E4 E5 E6 Number of Students 100 80 70 60 90 50 Tasks: a) Formulate the problem as a Constraint Satisfaction Problem (CSP) by clearly defining the variables, domains, and constraints. b) Use Backtracking with the Minimum Remaining Values (MRV) heuristic and Forward Checking to obtain a valid examination timetable. c) Show the domain of each variable after every assignment and provide the final timetable, ensuring that all student-conflict and hall-capacity constraints are satisfied.	
4	Game-Playing AI: You are developing an AI opponent for a board game. The current game position produces the following possible moves and terminal scores in the supplied game tree. Determine the move the AI should make by applying Minimax. Then apply Alpha-Beta Pruning and identify the branches that do not need to be evaluated. Explain how pruning improves the efficiency of the game-playing agent.	
	Game Playing, Minimax, Alpha-Beta Pruning	



5

Delivery Route Optimization:

A logistics company wants to minimize the total distance travelled by a delivery vehicle visiting multiple customers. An initial route and several neighboring routes are provided. Design an optimization strategy to improve the route. Apply the strategy step-by-step to the given routes, determine the final route, explain whether it represents a global optimum, and suggest how the algorithm could escape a local optimum.

Distance matrix (in km)					
	D	A	B	C	E
D	-	6	9	8	7
A	6	-	3	5	6
B	9	3	-	4	5
C	8	5	4	-	3
E	7	6	5	3	-

Some neighboring routes of the initial solution and their costs	
S1: D → A → C → B → E → D	= 26 km
S2: D → B → A → C → E → D	= 30 km
S3: D → A → B → E → C → D	= 24 km
S4: D → E → C → B → A → D	= 27 km
S5: D → A → E → B → C → D	= 23 km
S6: D → C → B → A → E → D	= 29 km

From S5, the best neighbor has cost 22 km.
From that solution, all neighbors have cost ≥ 22 km.
Apply the chosen local search strategy and report the final route.

Local Search, Optimization, Hill Climbing

Q1. SMART AMBULANCE NAVIGATION — A*

SEARCH

Given

Start: S, $h(S)=8$

Goal: G, $h(G)=0$

Heuristic values:

Node	$h(n)$
S	8
A	7
B	6
C	2
D	6
E	3
F	1
G	0

Formula

For A*:

$$f(n)=g(n)+h(n)$$

where:

$g(n)$ = actual cost from S to node

$h(n)$ = estimated cost from node to G

$f(n)$ = total estimated cost

Step 1 — Start at S

$$g(S)=0$$

$$h(S)=8$$

$$f(S)=0+8= 8$$

From S:

$$S \rightarrow A = 2$$

$$S \rightarrow D = 6$$

Calculate:

A

- $g(A)=2$
- $h(A)=7$
- $f(A)=2+7=9$

D

- $g(D)=6$
- $h(D)=6$
- $f(D)=6+6=12$

OPEN:

Node	g	h	f
A	2	7	9
D	6	6	12

Choose A because it has the lowest f.

Step 2 — Expand A

Connections from A:

- $A \rightarrow B = 4$
- $A \rightarrow D = 6$
- $A \rightarrow E = 4$

B

- $g(B)=g(A)+4=2+4=6$

- $h(B)=6$
- $f(B)=6+6=1$

D

Through A:

$$g(D)=2+6=8$$

But we already have:

$$g(D)=6$$

So keep the better value:

$$g(D)=6, f(D)=12$$

E

- $g(E)=2+4=6$
- $h(E)=3$
- $f(E)=6+3=9$

OPEN:

Node	g	h	f
E	6	3	9
B	6	6	12
D	6	6	12

Choose E.

Step 3 — Expand E

Connections:

- $E \rightarrow B = 2$
- $E \rightarrow D = 3$
- $E \rightarrow F = 4$

B through E

$$g(B)=6+2=8$$

But previous:

$$g(B)=6$$

Keep:

$$g(B)=6, f(B)=12$$

D through E

$$g(D)=6+3=9$$

Previous:

$$g(D)=6$$

Keep previous value.

F

- $g(F)=6+4=10$
- $h(F)=1$
- $f(F)=10+1= 11$

OPEN:

Node	g	h	f
F	10	1	11
B	6	6	12
D	6	6	12

Choose F.

Step 4 — Expand F

F connects to G.

$$g(G)=g(F)+2$$

The graph shows the F–G connection; using its road cost of 2:

- $g(G)=10+2=12$
- $h(G)=0$

Therefore:

- $f(G)=12+0=12$

OPEN now contains:

Node	g	h	f
B	6	6	12
D	6	6	12
G	12	0	12

There is a tie at $f=12$. We should not stop at G yet if another node with the same f may produce a cheaper goal path.

Choose B.

Step 5 — Expand B

B connects to C and G.

C

- $g(C)=g(B)+3$
- $g(C)=6+3=9$
- $h(C)=2$
- $f(C)=9+2=11$

G through B

$$g(G)=6+6=12$$

Same as the current value.

OPEN:

Node	g	h	f
C	9	2	11
D	6	6	12
G	12	0	12

Choose C.

Step 6 — Expand C

$C \rightarrow G$ has cost 2.

- $g(G)=g(C)+2$
- $g(G)=9+2=11$
- $h(G)=0$

Therefore:

$$f(G)=11+0=11$$

This is better than the previous $g(G)=12$.

Step 7 — Goal Reached

The final route is:

$$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$$

Total cost

$$2+4+3+2=11$$

Final A* Table

Expanded	$g(n)$	$h(n)$	$f(n)$
S	0	8	8
A	2	7	9
E	6	3	9
F	10	1	11
B	6	6	12
C	9	2	11
G	11	0	11

Final Answer

Expansion order:

$$S \rightarrow A \rightarrow E \rightarrow F \rightarrow B \rightarrow C \rightarrow G$$

Selected optimal route:

$$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$$

Optimal cost: 11

The route is optimal assuming the supplied heuristic is admissible/consistent and using the given edge costs.

Q2. AUTONOMOUS WAREHOUSE ROBOT — ONLINE SEARCH

The coordinates are:

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

S	Charging Station (Start)
G	Target Shelf (Goal)
#	Obstacle (Unknown initially)
?	Unknown cell

Known obstacles shown in the question

(2,4),(5,4),(4,3),(7,3),(2,2),(6,2),(3,1)

Step 1 — Start

Robot starts at: (1,5)

Goal: (7,1)

The robot repeatedly performs:

Observe→Plan→Act→Update→Replan

Step 2 — Move Right

The robot observes the neighbouring cells.

Move:

$$(1,5) \rightarrow (2,5)$$

Cell (2,5) is free.

Step 3

Move:

$$(2,5) \rightarrow (3,5)$$

Cell (3,5) is free.

Step 4

Move:

$$(3,5) \rightarrow (4,5)$$

Cell (4,5) is free

Step 5

Move:

$$(4,5) \rightarrow (5,5)$$

Cell (5,5) is free.

Step 6

Move:

$$(5,5) \rightarrow (6,5)$$

Cell (6,5) is free.

Step 7

Move:

$$(6,5) \rightarrow (7,5)$$

Cell (7,5) is free.

Step 8 — Move Up

Robot moves:

$$(7,5) \rightarrow (7,4)$$

Cell (7,4) is free.

Step 9 — NEW OBSTACLE DISCOVERED

The robot wants to move:

$$(7,4) \rightarrow (7,3)$$

But:

$$(7,3) = \text{Obstacle}$$

The robot did not know this initially.

Therefore it does not move into the obstacle.

It updates its internal map:

$(7,3) = \text{BLOCKED}$

This is the key online-search behaviour.

Step 10 — Replan

The original planned direction was upward.

But the second upward movement is blocked.

The robot searches for another route.

New route: $(7,4) \rightarrow (6,4)$

Step 11

Move: $(6,4) \rightarrow (6,3)$

Step 12

Move: $(6,3) \rightarrow (5,3)$

Step 13

Move: $(5,3) \rightarrow (5,2)$

Step 14

Move: $(5,2) \rightarrow (5,1)$

Step 15

Move: $(5,1) \rightarrow (6,1)$

Step 16 — Goal

Move: $(6,1) \rightarrow (7,1)$

The robot reaches: G

Final Robot Path

The complete path is:

$(1,5) \rightarrow (2,5) \rightarrow (3,5) \rightarrow (4,5) \rightarrow (5,5) \rightarrow (6,5) \rightarrow (7,5) \rightarrow (7,4)$

Obstacle discovered at: $(7,3)$

Compact Execution Table

Step	Position	Decision
0	(1,5)	Start
1	(2,5)	Move Right
2	(3,5)	Move Right

3	(4,5)	Move Right
4	(5,5)	Move Right
5	(6,5)	Move Right
6	(7,5)	Move Right
7	(7,4)	Move Up
8	(7,3)	Obstacle discovered
9	(6,4)	Replan
10	(6,3)	Move Up
11	(5,3)	Move Left
12	(5,2)	Move Up
13	(5,1)	Move Up
14	(6,1)	Move Right
15	(7,1)	Goal

Q3 — UNIVERSITY EXAMINATION SCHEDULING

Given Information

Examinations

Exam	Students
E1	100
E2	80
E3	70
E4	60

E5	90
E6	50

Halls

Hall	Capacity
A	120
B	90
C	60

Available time slots

T1, T2, T3

Conflict constraints

1. E1 and E2 have common students.
2. E1 and E3 have common students.
3. E2 and E4 have common students.
4. E3 and E5 have common students.
5. E4 and E5 have common students.
6. E2 and E6 have common students.

Step 1 — Define CSP

Variables

- $X = \{E1, E2, E3, E4, E5, E6\}$

Each variable represents an examination.

Domain

Initially:

$$D(E1)=D(E2)=\dots=D(E6)=\{T1,T2,T3\}$$

Constraints

Conflicting examinations cannot have the same time slot.

There is also a hall-capacity constraint:

$$\text{Capacity}(\text{Hall}) \geq \text{Students}(\text{Exam})$$

Step 2 — Determine Hall Domains

Based on capacity:

$$E1 = 100 \text{ students}$$

Only Hall A:

$$D_H(E1)=\{A\}$$

$$E2 = 80$$

$$D_H(E2)=\{A,B\}$$

$$E3 = 70$$

$$D_H(E3)=\{A,B\}$$

$$E4 = 60$$

$$D_H(E4)=\{A,B,C\}$$

$$E5 = 90$$

$$D_H(E5)=\{A,B\}$$

$$E6 = 50$$

$$D_H(E6)=\{A,B,C\}$$

Step 3 — Apply MRV

Initially every examination has three time-slot choices, so there is a tie.

Choosing E2 because it has the largest number of conflict relationships:

$$E2 \leftrightarrow E1, E4, E6$$

Assign:

$$E2=T2$$

Step 4 — Forward Checking

Since E1, E4 and E6 conflict with E2, they cannot use T2.

Therefore:

- $D(E1) = \{T1, T3\}$
- $D(E4) = \{T1, T3\}$
- $D(E6) = \{T1, T3\}$

For E3 and E5:

- $D(E3) = \{T1, T2, T3\}$
- $D(E5) = \{T1, T2, T3\}$

Step 5 — Assign E1

Choose:

$$E1 = T1$$

Because E1 conflicts with E2 and E3:

$$D(E3) = \{T2, T3\}$$

E2 is already T2, so no change.

Step 6 — Assign E4

Choose:

$$E4=T1$$

E4 conflicts with E2 and E5.

Therefore:

$$D(E5)=\{T2,T3\}$$

Step 7 — Assign E6

E6 has:

$$D(E6)=\{T1,T3\}$$

Choose:

$$E6=T1$$

E6 conflicts only with E2, and E2 is already T2, so this is valid.

Step 8 — Assign E3

E3 has:

$$D(E3)=\{T2,T3\}$$

Choose:

$$E3=T2$$

E3 conflicts with E1 and E5.

E1 is T1, so there is no conflict.

Therefore E5 cannot use T2:

$$D(E5)=\{T1,T3\}$$

Step 9 — Assign E5

Choose:

$$E5=T3$$

All constraints are satisfied.

Final Timetable

Exam	Students	Time	Hall
E1	100	T1	Hall A
E2	80	T2	Hall A
E3	70	T2	Hall B
E4	60	T1	Hall B
E5	90	T3	Hall A
E6	50	T1	Hall C

Capacity check

T1:

- $E1 \rightarrow \text{Hall A} \rightarrow 100/120$

- $E4 \rightarrow \text{Hall B} \rightarrow 60/90$
- $E6 \rightarrow \text{Hall C} \rightarrow 50/60$

T2:

- $E2 \rightarrow \text{Hall A} \rightarrow 80/120$
- $E3 \rightarrow \text{Hall B} \rightarrow 70/90$

T3:

- $E5 \rightarrow \text{Hall A} \rightarrow 90/120$

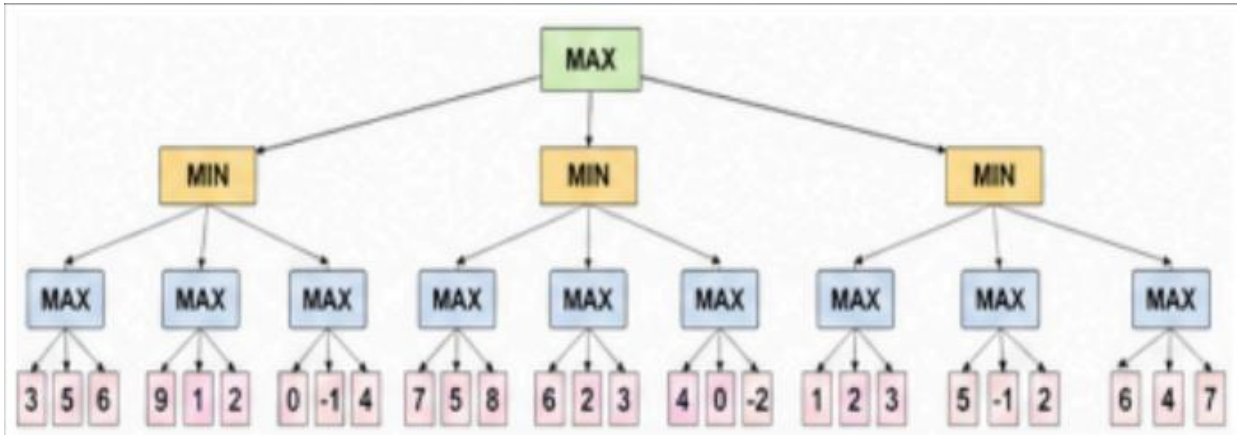
Everything fits.

Final answer

$$E1=T1, E2=T2, E3=T2, E4=T1, E5=T3, E6=T1$$

Q4 — GAME-PLAYING AI

The tree contains:



More precisely, the three MAX nodes have:

- MAX1 $\rightarrow$ 3, 5, 6
- MAX2 $\rightarrow$ 9, 1, 2
- MAX3 $\rightarrow$ 0, -1, 4

Middle MIN branch

- MAX4 $\rightarrow$ 7, 5, 8
- MAX5 $\rightarrow$ 6, 2, 3
- MAX6 $\rightarrow$ 4, 0, -2

Right MIN branch

- $\text{MAX7} \rightarrow 1, 2, 3$
- $\text{MAX8} \rightarrow 5, -1, 2$
- $\text{MAX9} \rightarrow 6, 4, 7$

Step 1 — Minimax: Left Branch

MAX1

$$\text{MAX}(3, 5, 6) = 6$$

MAX2

$$\text{MAX}(9, 1, 2) = 9$$

MAX3

$$\text{MAX}(0, -1, 4) = 4$$

Now MIN chooses the smallest:

$$\text{MIN}(6, 9, 4) = 4$$

So:

$$\text{Left MIN} = 4$$

Step 2 — Minimax: Middle Branch

MAX4

$$\text{MAX}(7,5,8)= 8$$

MAX5

$$\text{MAX}(6,2,3)= 6$$

MAX6

$$\text{MAX}(4,0,-2)= 4$$

MIN chooses:

$$\text{MIN}(8,6,4)= 4$$

Therefore:

Middle MIN=4

Step 3 — Minimax: Right Branch

MAX7

$$\text{MAX}(1,2,3)= 3$$

MAX8

$$\text{MAX}(5,-1,2)= 5$$

MAX9

$$\text{MAX}(6,4,7)= 7$$

MIN chooses:

$$\text{MIN}(3,5,7) = 3$$

Therefore:

$$\text{Right MIN} = 3$$

Step 4 — Root MAX

The three values are:

- Left = 4
- Middle = 4
- Right = 3

MAX chooses the highest:

$$\text{MAX}(4,4,3) = 4$$

Final Minimax Answer

$$\text{MAX} = 4$$

MAX can choose either the Left or Middle branch, since both have value 4.

Step 5 — Alpha-Beta Pruning

We evaluate from left to right.

Left branch

- All three MAX nodes are evaluated.
- Values: 6, 9, 4
- Therefore: $\text{MIN}=4$
- At root: $\alpha =4$

Middle branch

- First MAX: $\text{MAX}(7,5,8)=8$
- So: $\beta =8$
- Since: $\alpha =4 < \beta =8$
- continue.
- Second MAX: $\text{MAX}(6,2,3)=6$
- Now: $\beta =6$
- Still: $4 < 6$
- Continue.
- Third MAX: $\text{MAX}(4,0,-2)=4$
- Middle MIN = 4.

- No pruning occurs here.

Step 6 — Right Branch

- First MAX: $\text{MAX}(1,2,3)=3$
- Therefore: $\beta = 3$
- But root already has: $\alpha = 4$
- Now: $\alpha \geq \beta$
- $4 \geq 3$
- Therefore, the remaining two MAX nodes are pruned.
- Pruned branches
- $\text{MAX8} \rightarrow 5, -1, 2$
- $\text{MAX9} \rightarrow 6, 4, 7$
- So 6 leaf nodes are not evaluated.

Results

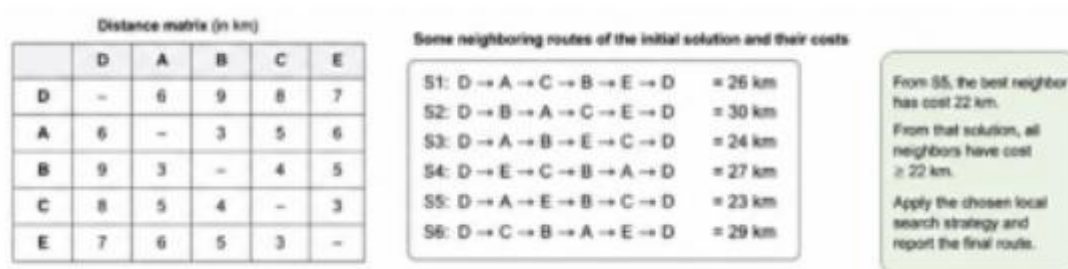
Method	Leaf Nodes Evaluated
Minimax	27
Alpha-Beta	21
Pruned	6

Final value : 4

Best move : Left or Middle

Alpha-Beta saving : $27 - 21 = 6$ nodes

Q5 — DELIVERY ROUTE OPTIMIZATION



Step 1 — Initial Solution

The initial route is: S0

with cost: 26 km

Step 2 — Generate Neighbors

The question provides six neighboring routes:

Solution	Cost
S1	26 km
S2	30 km
S3	24 km

S4	27 km
S5	23 km
S6	29 km

The best neighbor is the one with the smallest cost.

Therefore:

$\text{MIN}(26,30,24,27,23,29) = 23$

So select: S5

Step 3 — Move to S5

Current solution:

$S_0 = 26 \text{ km}$

New solution:

$S_5 = 23 \text{ km}$

Improvement:

$26 - 23 = 3 \text{ km}$

So Hill Climbing accepts S5.

Step 4 — Generate Neighbors of S5

According to the question:

From that solution, all neighbors have cost ≥ 27 km.

Therefore:

27, 28, 29,...

All of them are worse than: 23 km

Since this is minimization:

$$23 < 27$$

No neighboring solution improves S5.

Step 5 — Stop

Hill Climbing stops because there is no better neighboring solution.

Therefore:

$$S5 = 23 \text{ km}$$

is the final solution found by the algorithm.

Step 6 — Is It a Global Optimum?

No guarantee.

Hill Climbing only examines neighboring states.

It has reached a state where:

- all neighbours ≥ 27

while the current solution is:

- 23 km

Therefore S5 is a local optimum.

It may or may not be the global optimum.

Step 7 — How to Escape Local Optimum?

Possible methods:

Random Restart Hill Climbing

- Start from another initial route and perform hill climbing again.

Simulated Annealing

- Occasionally accept a worse move to escape local minima.

Stochastic Hill Climbing

- Randomly select among improving moves instead of always choosing the same one.

Conclusion

The Hill Climbing algorithm improves the initial solution from 26 km to 23 km. Since every neighbor of S5 has a cost of at least 27 km, the algorithm stops at S5. This is a local optimum, not necessarily a global optimum.

EVALUATION RUBRICS

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
CSP Problem Formulation		20% 4% – Clearly defines all variables, domains, and problem objectives correctly.	3% – Defines most variables and domains with minor errors.	2% – Partial formulation with some missing components.	0–1% – Incorrect or incomplete formulation.
Constraint Identification & Representation		6% 4% – Accurately identifies all student-conflict and hall-capacity constraints and represents them correctly.	3% – Identifies most constraints with minor errors.	2% – Identifies some constraints but misses important relationships.	0–1% – Constraints are largely incorrect or missing.
Application of Backtracking, MRV & Forward Checking		6% 6% – Correctly applies Backtracking, MRV, and Forward Checking with clear and logical steps.	4.5% – Applies the techniques correctly with minor mistakes.	3% – Partial application with unclear or inconsistent steps.	0–1.5% – Incorrect or inappropriate application of CSP techniques.

Step-by-Step Execution & Domain Updates		20% 4% – Provides a complete, logical execution trace and correctly updates domains after every assignment.	3% – Mostly correct execution with minor gaps.	2% – Provides limited steps with some incorrect domain updates.	0–1% – Incomplete or incorrect execution.
---	--	---	--	---	---

Final Timetable, Accuracy & Justification		% 2% – Produces a valid conflict-free timetable satisfying all constraints and capacity requirements with clear justification.	1.5% – Valid timetable with minor errors in justification.	1% – Partially valid timetable with some constraint violations.	0–0.5% – Invalid or missing timetable.
---	--	--	--	---	--